

An Experimental Proposal for Identifying Luminal Membrane-Tethering Proteins for Tau Filaments in Alzheimer's Brain-Derived Extracellular Vesicles

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Abstract

Extracellular vesicles propagate Alzheimer's disease tau pathology, and pathogenic truncated tau filaments inside these vesicles are physically tethered to the luminal membrane face by unidentified proteins. This proposal aims to identify this tethering machinery and thereby halt tau dissemination mediated by extracellular vesicles. We establish a tag-free cellular cargo model that generates native truncated tau vesicles, admitted into screening exclusively through a cryo-electron tomography structural checkpoint (Gate-0). A dual-track chemical-biology library of enzyme inhibitors and cell-permeable interfering peptides is screened across orthogonal membrane stripping and a multidimensional biochemical and seeding filter. Top candidates undergo endogenous knock-in validation by genome editing, 2×2 pharmacological occlusion, statistical correlative light and electron microscopy scoring of more than 1,000 vesicles, and in vivo propagation tests in mice. This pipeline converts a structural observation into a druggable target upstream of tau secretion.

Keywords: Alzheimer's disease, Tau filament tethering, Chemical biology screening

1. Introduction

1.1 AD & Tau Pathology

Alzheimer's disease (AD) is characterized by extracellular amyloid- β plaques and intracellular neurofibrillary tangles (NFTs) of hyperphosphorylated tau [1,2]. NFT burden and topography, rather than plaque load, track cognitive decline [3,4]. Hyperphosphorylation and truncation then drive tau aggregation into β -sheet filaments whose protease-resistant core is the microtubule-binding repeat region (MTBR) [5,6]. NFTs advance through connected brain networks, indicating prion-like intercellular propagation: misfolded tau released from donor neurons is taken up by connected recipients and recruits native tau through templated misfolding, a self-amplifying cascade demonstrated by inoculation and network-spread experiments [7,8]. As a cytosolic protein without a signal peptide, tau reaches the extracellular space through unconventional routes, including endo-lysosomal secretion [9–11]. Vesicle-mediated transfer is of particular interest, because membranous packaging shields tau seeds and directs their intercellular delivery [9].

1.2 EV Biogenesis

EVs are lipid-bilayer particles released as MVB-derived exosomes or plasma-membrane ectosomes [12–14], with ESCRT and lipid-driven machineries sorting cargo into intraluminal vesicles [13, 15]. Brain EVs carry tau seeds: mouse-brain small EVs seed aggregation threshold-dependently [16]; as little as 300 pg of AD brain EV tau induces widespread hippocampal pathology in wild-type mice, far exceeding equal doses of free tau [17]; and

seeding potency mirrors disease-specific tau conformations [18]. After HSPG-dependent uptake and macropinocytosis [19], internalized EVs permeabilize endolysosomal membranes, the rate-limiting gateway to cytosolic seeding, and fibrils themselves inflict nanoscale lysosomal damage [20–22]. How filaments are instead retained intact within the EV lumen, and by what tethering machinery, remains unknown [23].

1.3 Tau Filaments within EVs

AD paired helical and straight filaments share an invariant C-shaped protofilament fold, the "Alzheimer fold" (ordered core S305–R379) [24, 25], decorated with post-translational modifications [26], with a protease-resistant core matching the ~12-kDa fragment first isolated from PHFs [27]. These structures, however, came from detergent-extracted filaments and could not address EV packaging. Fowler et al. resolved this question [23]: AD brain EV tau resides in endo-lysosomal fractions as truncated, hyperphosphorylated, sarkosyl-insoluble ~12/24-kDa species, and cryo-ET reveals short PHF-like (~88%) and SF-like (~12%) filaments enclosed within large (>300 nm) vesicles (3 to >50 per vesicle, protease-protected), with EV-derived PHFs retaining the Alzheimer fold plus extra anionic densities at R349/K375. Most strikingly, at least one filament per EV is tethered, exclusively at its end, to the luminal membrane face via flexible, membrane-traversing densities that withstand pH-11 extraction, while the remaining filaments form lateral networks around the tethered one; none floats free [23]. End-restricted tethering implies that exposed terminal motifs engage an unidentified machinery, adaptors such as BIN1 or direct anchors, whose identity is entirely unknown and constitutes the target of this proposal [23].

2. Research Questions

2.1 Core Research Questions

Building on the tethering phenomenon revealed by Fowler et al. [23] (Section 1.3), this proposal addresses one primary question: **Which protein(s) physically tether the ends of truncated tau filaments to the luminal face of the limiting membrane of large (>300 nm) endo-lysosomal EVs in the AD brain?** Resolving it will also illuminate (i) the topological binding mode (transmembrane anchor, lipid-anchored molecule, or adaptor complex) and (ii) whether disrupting the tether de-anchors filaments and attenuates EV-mediated tau propagation in vivo.

2.2 Working Hypotheses

We discriminate among three candidate classes. Class I, tethering machinery: its loss frees the filaments (Simoa free/bound ratio $\uparrow$) without changing EV secretion and abolishes tether densities under CLEM/cryo-ET. Class II, formation/maintenance machinery: its loss additionally reduces large-EV abundance and total tau secretion. Class III, cargo-sorting machinery (ALIX, CHMP4, VPS4): its loss leaves EVs devoid of luminal filaments rather than de-anchored ones. All three classes can mimic "tether disappearance"; the readout chain of Gate-0 admission, dual stripping, the Simoa golden criterion, and occlusion distinguishes them one by one.

3. Methods

The study couples chemical-biology perturbation with multi-scale structural readouts in a six-step pipeline (Section 4; Appendix). HEK293T cells expressing tag-free P301S 1N4R tau are

inoculated with AD brain-derived PHF seeds [23]; hollow-fiber culture supplies $\sim 10^{12}$ EVs per batch, and only lines passing a cryo-ET structural checkpoint (Gate-0) enter screening. EV handling follows MISEV practice [14]: OptiPrep F4–F6 fractionation, >300 -nm tangential-flow cutoff, and proteinase K protection. Enzyme-inhibitor and cell-permeable PPI-peptide libraries are screened against dual membrane stripping (pH 11 versus neutral elution) and a three-dimensional filter — cytotoxicity counter-screening, dual-channel Simoa free/bound tau [28–30], an LC-MS/MS phospho-panel [31], and FRET/NanoBiT seeding [32,33] — with eight-point dose-response and three-tier mechanistic classification. Candidates are validated by siRNA deconvolution in iPSC-derived neurons [34], endogenous CRISPR knock-in, 2×2 pharmacological occlusion, a >1000 -EV CLEM/Tokuyasu structural verdict, and hippocampal injection into wild-type or PS19 mice with AT8 readout [17, 23]; every readout is pre-tied to an explicit decision criterion (Sections 4.7–5).

4. Experimental Design

4.1 Step 1: Building the Cargo Model

HEK293T cells are transfected with tag-free P301S 1N4R tau and inoculated with AD brain-derived PHF seeds from at least three donors [35] (Figure 1). Endogenous proteases truncate the seeded filaments and produce native cargo ($\text{TauC}^+/\text{MC1}^+/\text{PHF1}^-$) matching human brain EV tau [23]. Because tau is untagged, high-secretion clones are selected by MC1/AT8 immunofluorescence or FRET rather than YFP-based FACS. No genetic blockade of EV biogenesis is applied; production is scaled in hollow-fiber bioreactors, offsetting the $\sim 1/1000$ frequency of filament-bearing EVs by absolute particle number. Cargo release criteria are

defined in Appendix Table 2. Gate-0 admission then follows (Figure 2): before any screening, model F4–F6 EVs must show at least one end-tethered filament and no free-floating luminal filaments by cryo-ET or immuno-EM [23]. Failure returns the pipeline to seed, clone, and culture optimization.

4.2 Step 2: Dosing Window and Large-EV Isolation

Production-to-secretion kinetics are measured by pulse-chase or a CD63-pHluorin reporter (Figures 3, 4). Two release curves are acquired in parallel: total tau (Simoa) and seed-competent tau (FRET biosensor). The two pools exit cells by different routes, and seed-competent tau is exported preferentially by lysosomal exocytosis [36, 37]. The dosing window is therefore set on the seed-activity curve to cover MVB maturation, when tether assembly must occur, with a lysosomal-exocytosis inhibitor as a mechanistic control. Isolation and quality-control parameters are summarized in Appendix Table 3.

4.3 Step 3: Dual-Track Screening and Dual Membrane Stripping

Screen A comprises annotated inhibitors of modifying enzymes (Figure 5; Appendix Table 4). Screen B comprises cell-permeable interfering peptides against BIN1-SH3 and the ESCRT surfaces ALIX-Bro1, CHMP4, and VPS4 [13, 15, 38], so that a purely structural tether can be hit independently of Screen A. Peptides that block ESCRT globally suppress ILV formation and are removed by the secretion counter-screen (Appendix Table 1). Because BIN1 is cytosolic, its peptide must engage the cytosolic face during MVB formation [38]. The two stripping systems are complementary (Figure 6). System 1 (pH 11) matches a transmembrane or lipid-

anchored tether, fitting the membrane-traversing densities seen by tomography [23]; System 2 (neutral elution) preserves peripheral proteins. If the tether is a peripheral adaptor, pH 11 removes it in wild-type and knockout material alike, and only System 2 resolves bound from free tau. Because filaments form lateral networks inside the vesicle [23], stripping readouts are paired with proteinase K accessibility assays and sampling imaging.

4.4 Step 4: Hit Filtering and Three-Tier Classification

Hits are filtered by the golden criterion (Figure 7; Appendix Table 5): Simoa core-channel free ratio increased, N-terminal total constant, and seed activity at least 100% of control. Seeding is measured by Lipofectamine-assisted uptake into biosensor cells [23], which raises signal 100–1000-fold [32] and is declared to report total seeding rather than natural uptake [20]. Low-abundance samples are immunoprecipitated before application [32]. Surviving hits receive eight-point EC₅₀ curves and are classified by a three-tier tree (Figure 9): loss of the >300 nm subpopulation indicates Class II (formation/maintenance); reduced per-particle loading indicates Class III (cargo sorting); loaded but de-anchored tau indicates Class I (tethering). Hits affecting both size and de-anchoring are judged dual-function, with BIN1 as the prototype [38]. All deconvolution uses siRNA in iNeurons (Figure 8). If both screens fall silent, the pre-specified proteome-mining and lipidomics fallback arms activate.

4.5 Step 5: Endogenous Validation and Structural Verdict

Candidates are validated at the endogenous locus by CRISPR-Cas9 knock-in (Figure 10), with PRM proteomics calibrating expression to 0.5–2× wild-type levels. A 2×2 occlusion design

(wild-type/knockout $\times$ vehicle/hit) then tests target specificity. No additive de-anchoring on the knockout background confirms that drug and knockout act on the same protein. Separation-of-function rescue maps the interaction interface, using kinase-dead or phosphosite mutants for Screen A hits and domain mutants for Screen B hits. Rescue is accepted only when biochemical, structural, and functional criteria are met simultaneously. The final structural verdict is statistical (Figure 11): power-sized CLEM or Tokuyasu cryo-immunogold scoring covers more than 1000 large EVs, with EV size distribution as a second criterion distinguishing tether loss from container loss. Cryo-ET provides representative 3D display only. Class II candidates are judged by MVB/ILV morphometry and live-cell MVB dynamics, and Class III candidates by time-resolved cargo-sorting assays.

4.6 Step 6: In Vivo Propagation Test

EVs from tether-knockout and rescue lines are injected into the hippocampus of PS19 mice (Figure 12), following Fowler et al. [23]. Wild-type mice are an equally valid host, because 300 pg of AD brain EV tau induces spreading in wild-type animals, and EV-packaged tau seeds outperform equal doses of free fibrils [17]. The design rests on two established premises. Pathological seeding burden predicts tauopathy in vivo [39], and blockade of EV secretion halts propagation [40]. The central prediction is that untethering blocks EV-tau spreading. At 2–4.5 months, AT8 immunohistochemistry quantifies the spread area. Animals are male only or sex-stratified, because BIN1-linked effects are sex-dependent [38].

4.7 Control Experiments

Four control modules are integrated across the pipeline to guarantee cargo fidelity, eliminate false positives from global EV-biogenesis inhibition, and distinguish filament de-anchoring from upstream packaging defects (Appendix Table 1).

5. Expected Results

The pipeline is expected to deliver four quantitative milestones. First, Step 1 produces a Gate-0-passing cargo line: liter-scale batches ($\sim 10^{12}$ EVs) whose F4–F6 large EVs carry natively truncated, end-tethered filaments matching human AD brain EV cargo [23], demonstrating that tethering is cell-autonomous and screenable. Second, Steps 2–4 form a defined attrition funnel: about 500 compounds and peptides enter the screen; 10–15 hits are expected to survive the 3D matrix filter under the golden criterion; dose-response analysis reduces these to 3–5 candidates with EC_{50} values, each assigned a mechanistic class. We anticipate that genuine Class I hits exist, because the tether densities traverse the limiting membrane [23]; Class II and III hits are retained as a mechanism map of the packaging machinery. Third, Step 5 is expected to confirm one to two tethering proteins at endogenous expression: occlusion establishes target specificity, rescue maps the interaction domain, and CLEM scoring of more than 1000 large EVs delivers the statistical verdict. Fourth, Step 6 tests the central prediction in vivo: tether-knockout EVs should show significantly reduced AT8-positive spread, and rescue EVs should restore spreading, establishing tethering as a druggable node in tau propagation [17, 40].

Alternative outcomes are pre-interpreted. If both screens fall silent, the pre-specified fallback arms activate: nomination of 5–10 membrane proteins from the F4–F6 proteome [23], and F4–

F6 lipidomics for direct lipid anchoring. A null protein screen would itself shift the model toward non-protein anchoring. If untethering fails to block spreading in vivo, the tether likely serves cargo protection rather than release control, redirecting therapeutic focus toward the EV biogenesis step [13, 15].

6. Discussion

6.1 Innovation

This proposal converts the structural observation of end-tethered tau filaments in brain EVs [23] into the first systematic search for the tethering proteins. Three design elements distinguish it: phenotype-first Gate-0 admission, which prevents screening an irrelevant model; a pharmacological screen returning tool molecules with EC_{50} values, with 2×2 occlusion adding target validation unavailable to genetic screens; and a power-sized CLEM verdict (>1000 EVs) that turns electron microscopy from anecdote into statistics, while the three-tier classification tree converts off-target hits into a mechanism map.

6.2 Limitation

Three limitations remain. The HEK293T model is not neuronal and may miss neuron-specific tethers; siRNA deconvolution in iNeurons [34] and cross-validation against the human brain EV proteome [23] give such candidates two further chances of capture. The protein-centric screens would miss RNA- or lipid-mediated anchoring; the pre-specified lipidomic and data-driven fallback arms address this directly. Cultured-cell EV tau may also be hypophosphorylated relative to brain EV tau [41], so cargo decisions rest on the truncation profile rather than PHF1; and the transfection-assisted seeding assay measures total seeding

rather than natural uptake [20, 32], offset by the in vivo step, where mice receive native EVs without transfection reagent. The rarity of filament-bearing EVs is managed by scale-up and immunoprecipitation enrichment [32].

6.3 Significance

Identifying the tether completes the molecular itinerary of EV-mediated tau propagation, from luminal packaging [23] to intercellular spreading [17], and nominates a drug target upstream of secretion, a concept supported by the GWAS-implicated role of BIN1 in EV-mediated tau spreading [38] and by in vivo propagation blockade through EV-secretion inhibition [40]. Tether ectodomains may further extend the CSF tau-fragment biomarker framework [29, 30] to the packaging machinery, and the pipeline is portable to other EV-cargo anchoring problems.

6.4 Conclusion

The pipeline — phenotype-gated model, dual-track screen, quantitative filter, statistical structural verdict, in vivo test — is expected to deliver one to two validated tether proteins with a defined pharmacological handle. Whether the tether proves protein, lipid, or both, the result converts a cryo-ET observation into mechanistic and therapeutic leverage against tau spreading.

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Appendix

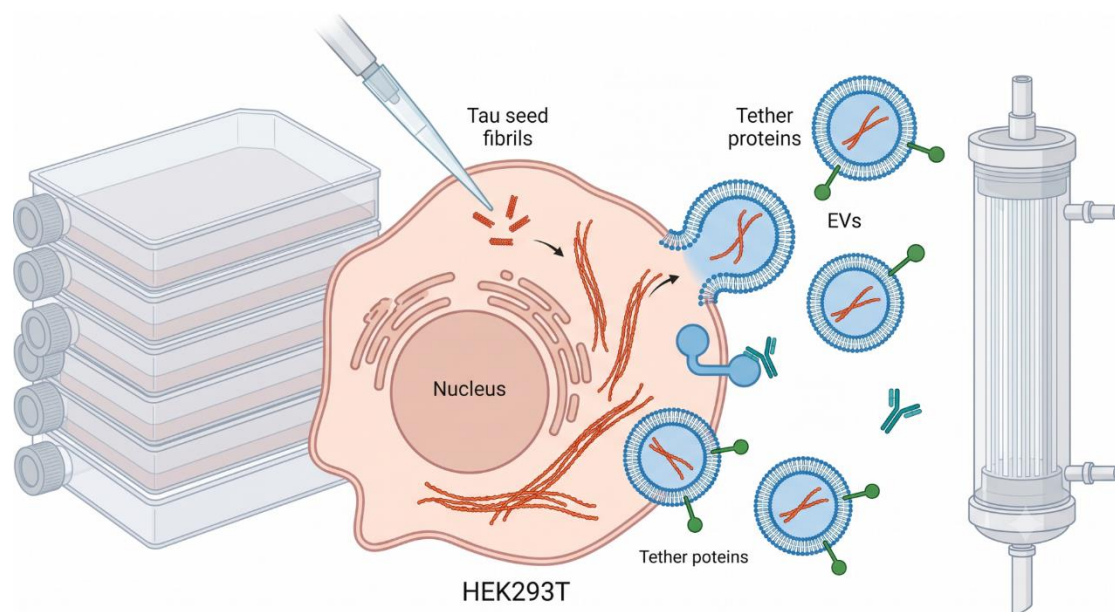


Figure 1 | Cargo model and hollow-fiber EV scale-up

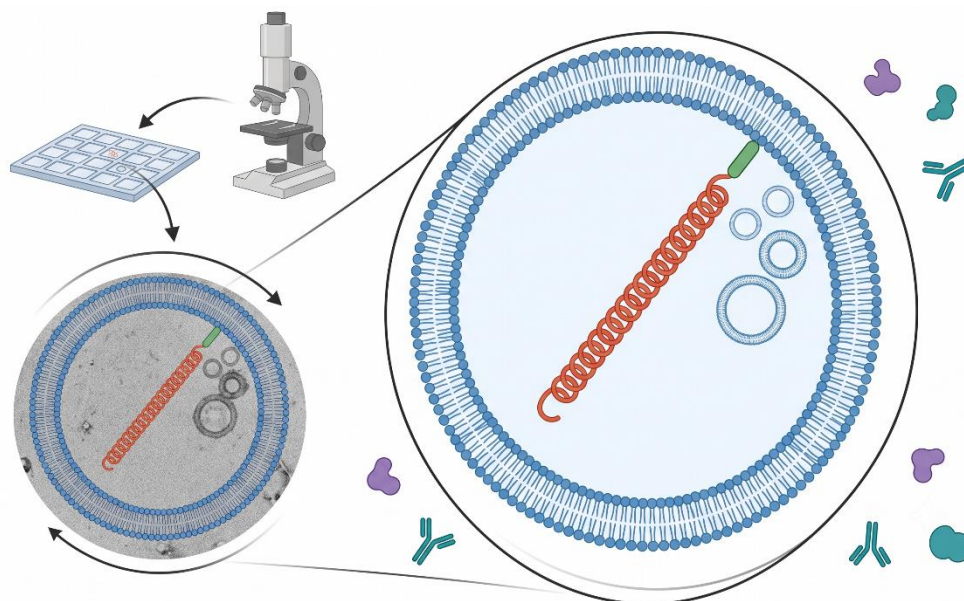


Figure 2 | Gate-0 structural admission by cryo-ET/CLEM

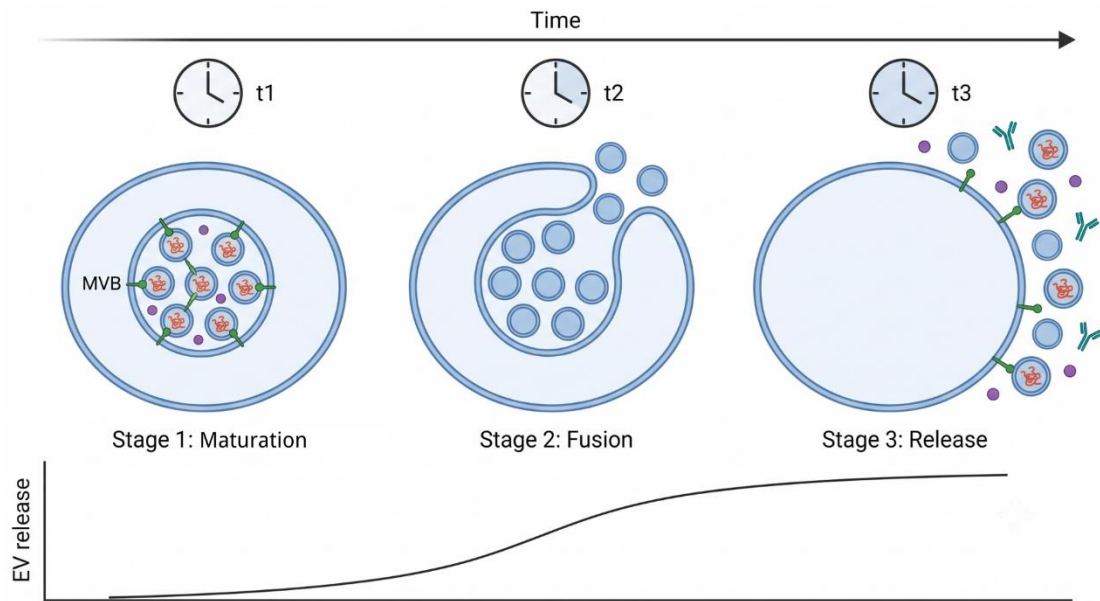


Figure 3 | EV biogenesis time course: maturation, fusion, release

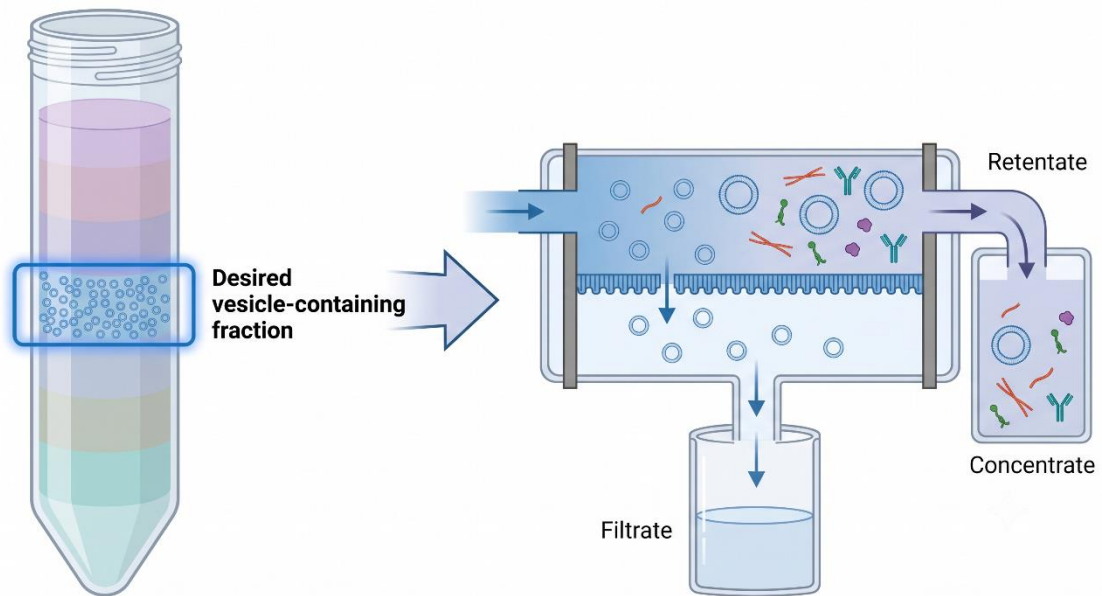


Figure 4 | OptiPrep density fractionation and tangential-flow concentration

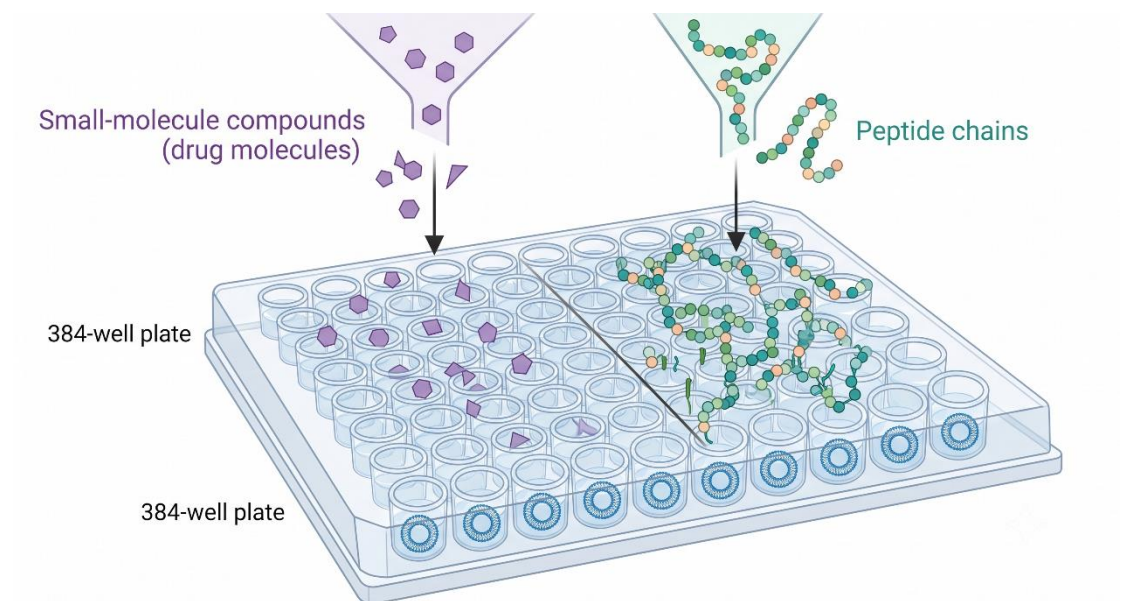


Figure 5 | Dual-track library screening in 384-well format

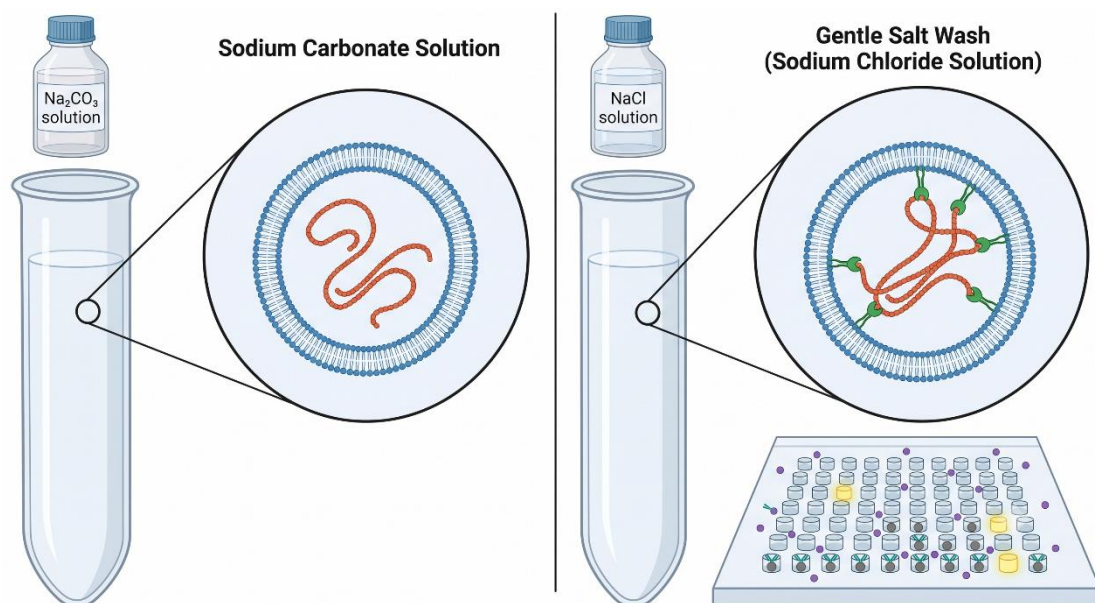


Figure 6 | Dual membrane stripping: pH 11 carbonate versus gentle NaCl wash

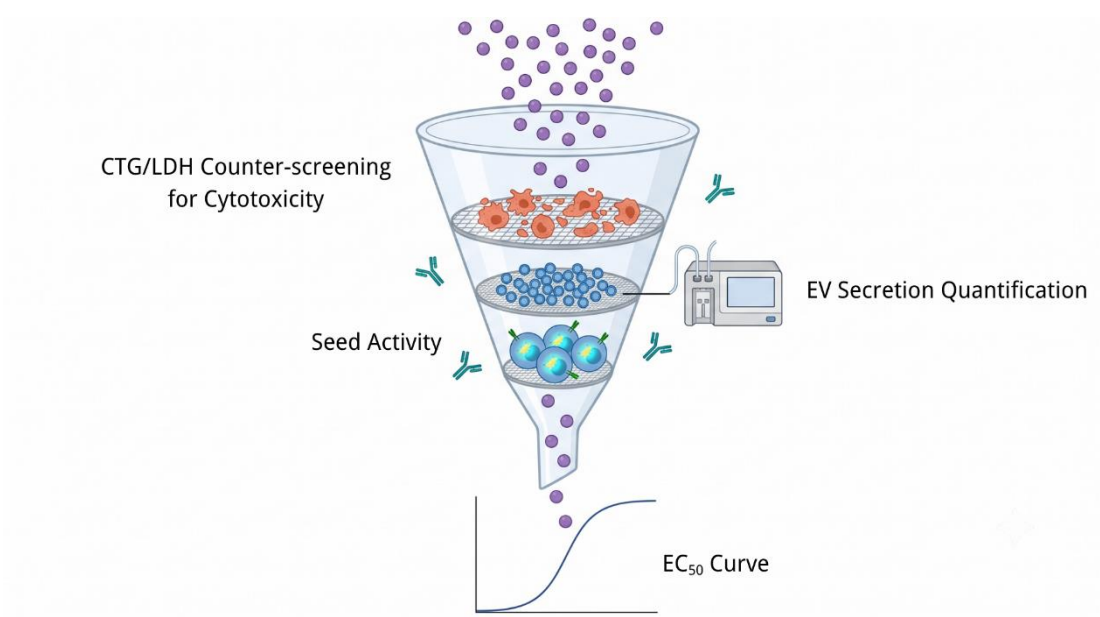


Figure 7 | Three-dimensional matrix filter and EC₅₀ confirmation funnel

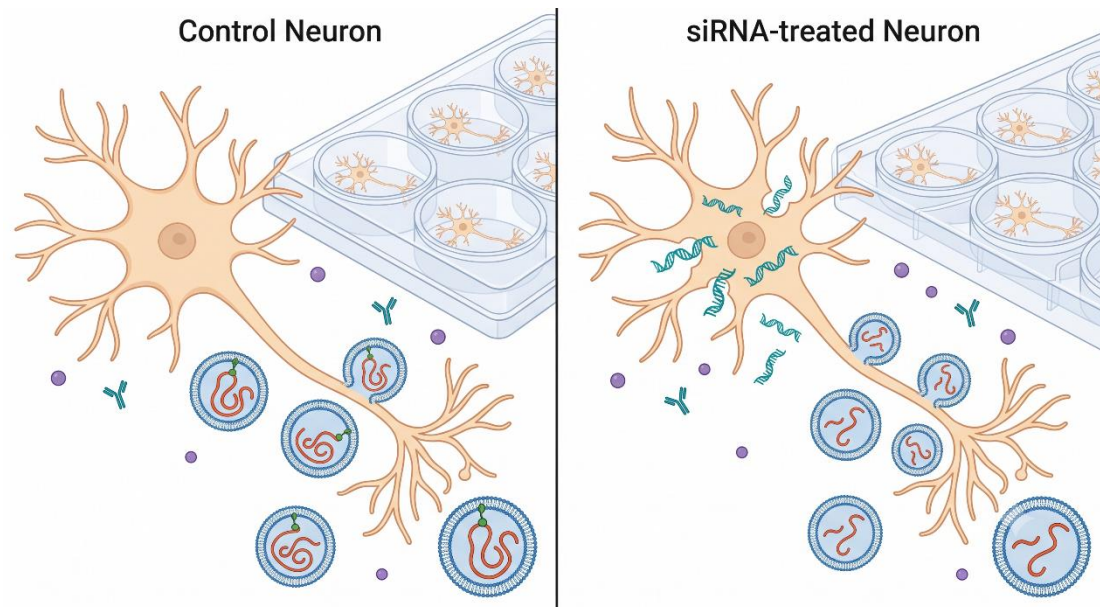


Figure 8 | siRNA validation of candidate tethers in iPSC-derived neurons

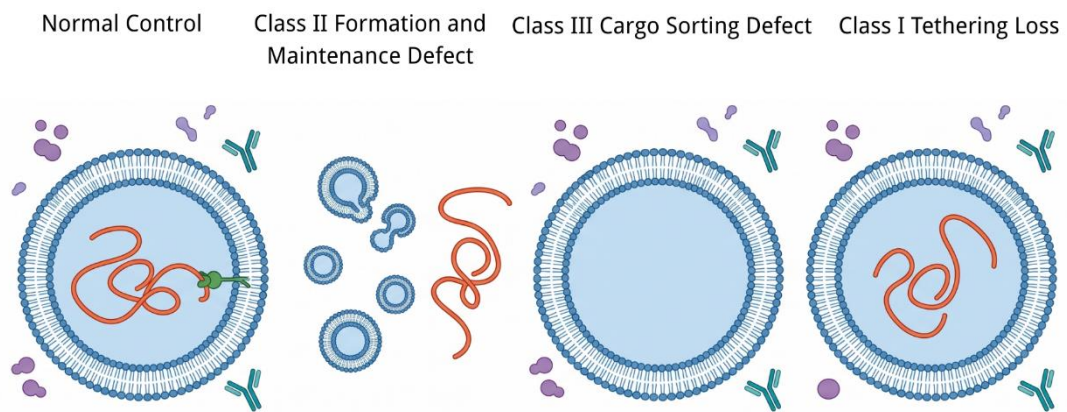


Figure 9 | Three-tier phenotypic classification of screening hits

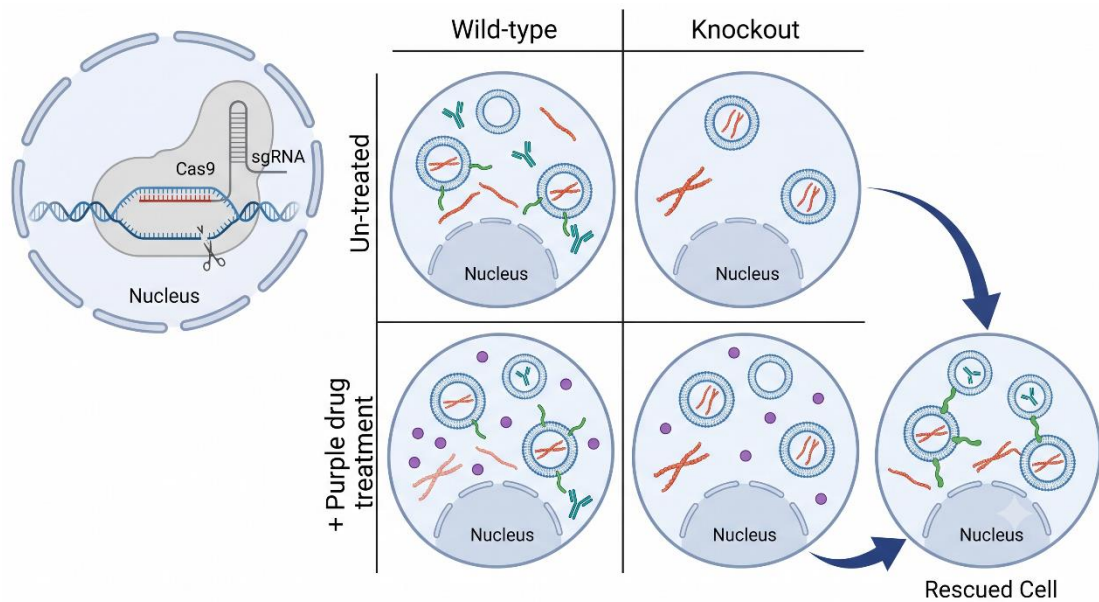


Figure 10 | CRISPR knockout, 2x2 pharmacological occlusion, and rescue

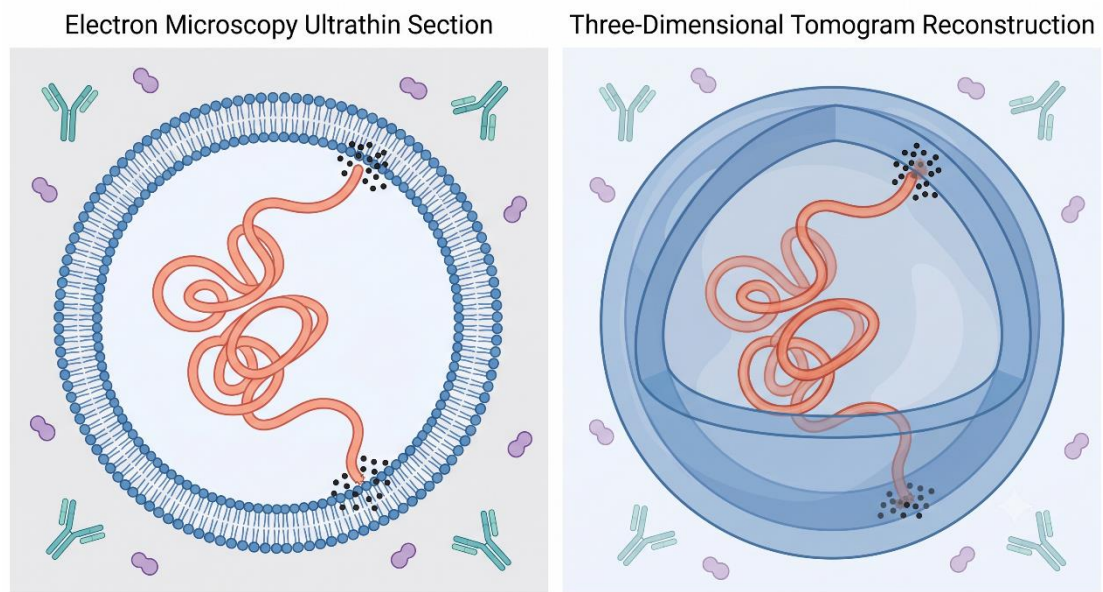


Figure 11 | Structural verdict: ultrathin-section immuno-EM and tomographic reconstruction

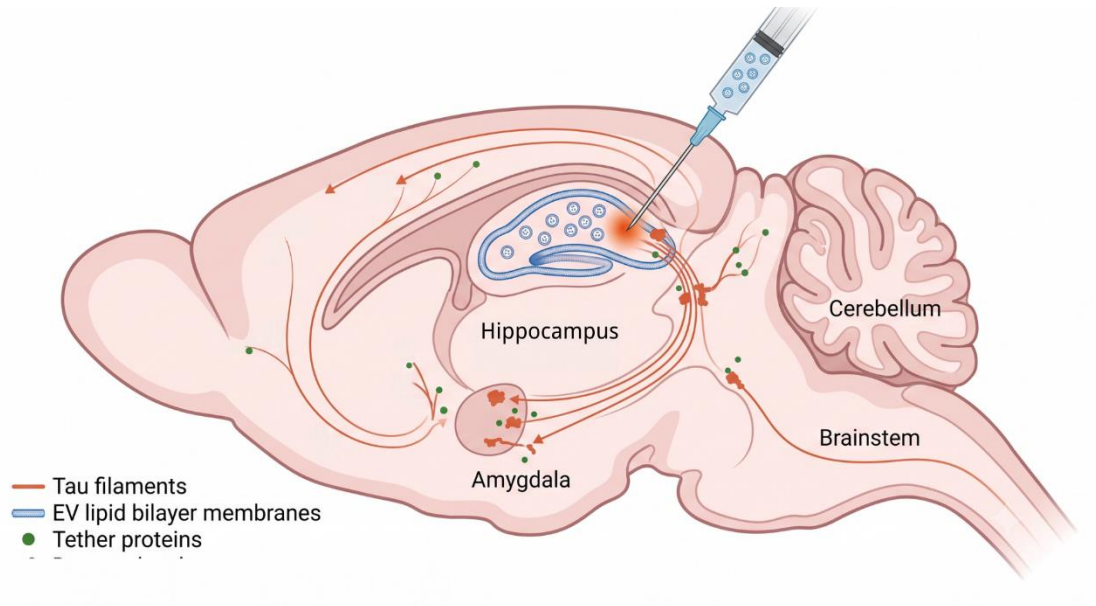


Figure 12 | In vivo propagation readout after hippocampal EV injection

Table 1 | Key Control Experiments and Quality Control (QC) Framework

Control Module	Experimental Methodology	Primary Scientific Objective
1. Truncated Tau Cargo Fidelity Control	Lysates of isolated F4–F6 EVs from the HEK293T model are analyzed by Western blotting using a targeted antibody panel (TauC ⁺ /Tau13 ⁻ /HT7 ⁻ /MC1 ⁺), sarkosyl insolubility assays, and antibody-free LC-MS/MS mapping of cleavage sites and phosphorylation profiles [23]. PHF1 serves as an auxiliary marker [23].	Confirm that the cellular model generates native truncated tau species matching post-mortem AD brain EV cargo (dominant ~12-kDa fragment, N-/C-terminally truncated, PHF1-negative), verifying that the identified tether possesses genuine pathological conservation [23].
2. OptiPrep Pre-fractionation & Parallel Stripping Control	Isolated F4–F6 fractions (11%–15% OptiPrep) are subjected in parallel to System 1 (0.1 M Na ₂ CO ₃ , pH 11 alkaline stripping) and System 2 (1 M NaCl or 0.05% digitonin neutral elution), coupled with proteinase K (PK) ± Triton X-100 protection assays [23].	Protect against screening failure if the tether is a peripheral membrane adaptor (e.g., BIN1-like proteins). While alkaline treatment hydrolyzes peripheral tethers in both WT and KO, neutral elution preserves peripheral binding to accurately resolve free vs. membrane-bound tau [23].
3. Dual-Track Screen A/B Parallel Control	Primary screening splits large EVs into Screen A (annotated small-molecule enzyme inhibitors) and Screen B (cell-permeable CPP-interfering peptides targeting BIN1-SH3, ALIX-Bro1, CHMP4, and VPS4) [23]. Hits are counter-screened against total EV secretion (NTA particle counts and CD81/TSG101/LAMP2 levels) [14].	Overcome the non-druggable pocket limitations of structural/scaffold tether proteins [23]. Counter-screening removes ESCRT-blocking false positives that suppress overall ILV/EV formation rather than specifically disrupting tethering [14].
4. EV Secretion & Packaging Normalization Control	Parallel quantification of NTA particle concentrations, tetraspanin/LAMP2 protein markers, and PK protection across all treated wells, normalizing de-anchoring readouts to "bound tau per EV particle" [14].	Distinguish filament de-anchoring (Class I) from upstream defects in EV biogenesis and cargo packaging (Class II/III), eliminating biogenesis-inhibition artifacts [14].

Table 2 Cargo Release Criteria and Gate-0 Admission		
Checkpoint	Method	Pass criterion
Seed batch quality control	Aggregation kinetics (lag phase, growth slope, plateau)	Consistent across ≥ 3 independent AD donors [35]
Sarkosyl insolubility	Detergent extraction and immunoblot	Insoluble filament pellet
Truncation profile	TauC/Tau13/HT7 immunoblot	TauC ⁺ /Tau13 ⁻ /HT7 ⁻ [23]
Dominant species	Immunoblot	~12-kDa band, PHF1-negative [23]
Conformation	MC1 immunoreactivity	MC1 ⁺
Seeding activity	HEK293-based tau seeding assay	Positive [32, 33]
PHF1 status	Auxiliary compositional marker only	Reported, not a release criterion [41]
Gate-0 (structural)	Cryo-ET or immuno-EM of F4–F6 EVs	≥ 1 end-tethered filament per EV; no free-floating luminal filaments [23]

Table 3 EV Isolation and Quality-Control Parameters		
Parameter	Setting	Rationale
Density fractionation	OptiPrep gradient, 11–15% interval (F4–F6)	LAMP2-enriched large-EV fractions [23]
Size cutoff	Tangential-flow filtration, >300 nm	Retains the large-EV subpopulation
Filtration	0.2- μ m filtration avoided; low-speed clarification only	Preserves >300 nm EVs
Capture strategy	CD9/CD63/CD81 phenotyping before immunocapture; mixed-tetraspanin capture or omission if CD81-low	CD81 concentrates in light fractions; tau ⁺ F4–F6 EVs are LAMP2-enriched [23]
Luminal enclosure	Proteinase K $\pm$ Triton X-100 protection assay	Confirms luminal cargo enclosure [23]

Table 4 Dual Membrane-Stripping Systems		
System	Condition	Target topology and logic
System 1 (harsh)	0.1 M Na ₂ CO ₃ , pH 11 alkaline stripping	Matches a transmembrane or lipid-anchored tether, consistent with membrane-traversing densities [23]
System 2 (gentle)	1 M NaCl wash or 0.05% digitonin neutral elution	Preserves peripheral adaptors; resolves bound from free tau when pH 11 removes the tether in all samples
Paired control	Proteinase K accessibility + sampling imaging	Excludes false membrane binding from lateral filament

Table 5 | Three-Dimensional Matrix Filter and Dose-Response

Dimension	Method	Role
Toxicity	CTG/LDH counter-screen	Excludes cytotoxic hits
Free/bound partition	Dual-channel Simoa: core channel (MTBR, ~residues 297–391) [28]; N-terminal channel (total tau) [29, 30]	Golden criterion: core free ratio ↑, N-terminal total constant
Phospho-panel	Antibody-free LC-MS/MS, seven sites [31]	Flags fragment-composition shifts
Seeding	FRET or plate-based NanoBiT biosensor, Lipofectamine-assisted [23, 32, 33]	Seed activity ≥100% of control
Responsiveness	Tau-siRNA well per plate [34]	Verifies knockdown responsiveness
Orthogonal seeding	PHF1 immunofluorescence	Nascent aggregates are pS396/pS404-positive [41]
Dose-response	Eight-point EC ₅₀ curves	Quantitative hit grading